



Society of Petroleum Engineers

SPE-228951-MS

From Data to Decision: The Role of 4.IR in Shaping the Future of Well Control and Integrity

Hany Gamal, Sherif Emam, and Suha Saif, Weatherford Saudi Arabia; Ahmed Alsabaa and Salaheldin Elkatatny, King Fahd University of Petroleum & Minerals

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/228859-MS](https://doi.org/10.2118/228859-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Drilling operations for oil and gas wells, constituting up to 40% of the total well cost, necessitate cutting-edge solutions for enhanced safety, efficiency, and well integrity. This research focuses on the integration of fourth industrial revolution (4.IR) technologies, specifically addressing well integrity within well control domains. The strategic application of machine learning, the internet of things, robotics, automation, and big data analytics aims to optimize operations, mitigate risks, and safeguard well integrity.

In the intricate landscape of drilling operations, the 4.IR technology becomes instrumental in ensuring well integrity. Rig sensors produce extensive, high-frequency data streams, and this research intricately examines how 4.IR technologies facilitate in-depth data analysis for safeguarding well integrity. The study delves into the technical advancements of 4.IR solutions, focusing on machine learning applications tailored for well integrity concerns, including early kick detection, pipe stuck prediction, wellbore stability, flux estimation, pore pressure prediction, and managed pressure drilling. Various machine learning techniques, data sources, features engineering, and target outputs achieved by developed models are explored.

The research highlights machine learning's outstanding impact on both the technical and economic aspects of well control and integrity. Case studies in the oil and gas industry demonstrate significant reductions in non-productive time (NPT), enhanced drilling performance, heightened safety, and lowered operational costs, with a central focus on preserving well integrity. The paper concludes by suggesting future avenues for machine learning applications, accentuating advanced data analysis, and addressing uncertainties in the well integrity domain. This collective insight positions machine learning as a transformative force in ensuring robust well control practices, aligning operational efficiency with economic viability while continually prioritizing the critical element of well integrity.

This research significantly contributes to the industry and academia by specifically addressing the role of 4.IR technologies in ensuring well integrity. Key features such as real-time monitoring, advisory systems, automation, digitalization, time and cost savings, and high accuracy for well integrity prediction and classification are explored. The integration of well integrity as a focal point within 4.IR applications in well control establishes a comprehensive framework for advancing safety and efficiency.

Key Words: Well control, 4.IR, Automation, Machine Learning, Applications Overview, Well Integrity, Technology Innovations

Introduction

Drilling operations in the oil and gas industry are inherently complex and capital-intensive, with well construction costs often constituting a significant portion, up to 40%, of the total well cost. Beyond the financial implications, these operations carry substantial risks related to safety, environmental impact, and asset integrity. Ensuring robust well control and integrity throughout the lifecycle of a well is paramount to prevent catastrophic events such as blowouts, uncontrolled fluid releases, and environmental contamination. Well control refers to the techniques and procedures used to maintain hydrostatic pressure in the wellbore to prevent the influx of formation fluids, while well integrity encompasses the technical, operational, and organizational solutions to reduce the risk of uncontrolled release of formation fluids throughout the life cycle of a well. The failure to maintain either can lead to severe consequences, including loss of life, significant financial penalties, environmental damage, and reputational harm to operating companies.

Challenges in Traditional Well Control and Integrity Management

Traditional approaches to well control and integrity management often rely on manual inspections, periodic maintenance schedules, and threshold-based alarm systems. These methods, while foundational, possess inherent limitations. They are typically labor-intensive, reactive rather than proactive, and may not capture the subtle, complex interactions that precede a critical failure. Furthermore, data across disparate systems hinders a holistic analysis of well conditions and prevents the early detection of potential issues. These limitations underscore the need for more advanced, integrated, and intelligent approaches to well integrity management.

Fourth Industrial Revolution (4.IR) Technologies

The fourth industrial revolution (4.IR), characterized by the fusion of digital, physical, and biological spheres, offers transformative potential for the oil and gas industry (Figure 1). Technologies such as machine learning (ML), the internet of things (IoT), robotics, automation, and big data analytics are poised to revolutionize how drilling operations are managed, risks are mitigated, and well integrity is safeguarded. These technologies enable unprecedented levels of data collection, real-time monitoring, predictive analytics, and autonomous decision-making, moving the industry from reactive problem-solving to proactive risk management.

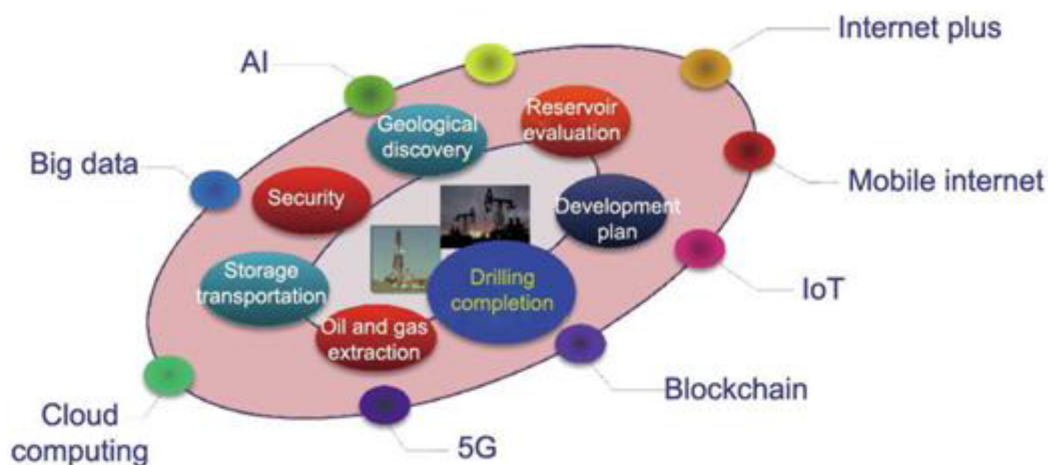


Figure 1—Leveraging AI technology across oil and gas [after (Li et al., 2022)]

This research focuses on the strategic integration of 4.IR technologies, specifically addressing well control domain. The primary objective is to explore how the application of machine learning, digital twins, internet of things, robotics, automation, and big data analytics can optimize drilling operations, mitigate risks, and enhance integrity. The study delves into the technical advancements of 4.IR solutions, focusing on academic simulator and machine learning applications tailored for critical well integrity concerns, including early kick detection, pipe stuck prediction, wellbore stability, flux estimation, pore pressure prediction, and managed pressure drilling technology innovation.

Digital Twins and Academic Simulators: Well Control From Virtual to Reality

Well control engineering demands precise training and predictive capabilities to manage high-risk drilling scenarios, and the integration of digital twin technology with academic drilling simulators creates a transformative framework for well control education and research by merging real-time operational data with immersive simulation (Sudhakar, 2023).

The digital twin functions as a dynamic virtual replica of a physical drilling asset, continuously updated via real-time sensor data (including downhole pressure, flow rates, and torque) or synthetic data while integrating physics-based models like multiphase flow hydraulics and kick dynamics to simulate well behavior under varying conditions (Figure 2). Complementing this, the academic drilling simulator provides a controlled environment with high-fidelity physics engines, realistic drill-floor interfaces, and scenario libraries covering kicks, losses, and equipment failures (Said et al., 2022). The integration methodology follows a closed-loop workflow: real-time rig data streams (standardized via WITSML) populate the digital twin, which then feeds live parameters into the drilling simulator; the drilling simulator runs scenarios using the digital twin's current wellbore state, physics models, and historical data, allowing trainees to respond to events like gas influx while the digital twin validates outcomes.

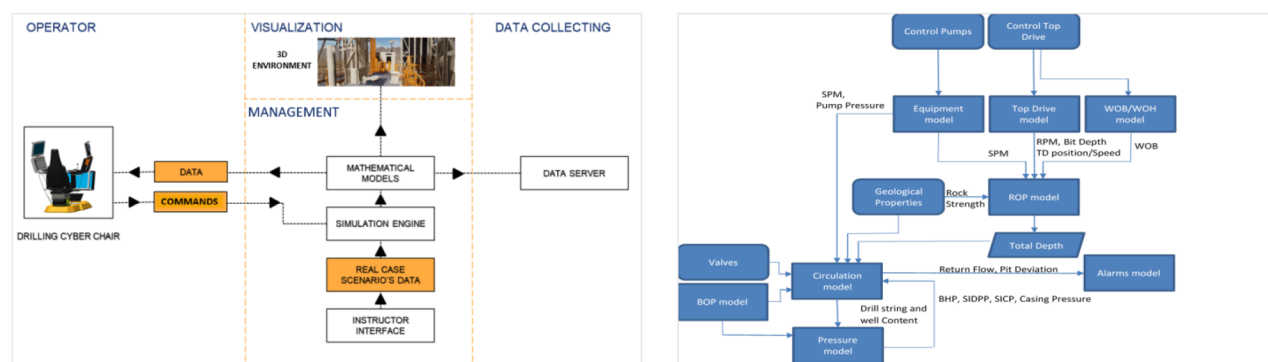


Figure 2—Digital Twin Simulator Workflow [after (Gallerati et al., 2023)]

The digital twin's engineering models and/or machine learning algorithms forecast pressure trends or kick evolution, visualized within the drilling simulator's interface (Figure 3) to enable proactive decision-making; and data from drilling simulators experiments refines the digital twin's physics models, enhancing accuracy for both training and real-world monitoring.



Figure 3—Drilling Parameters Visualization for Drilling Simulator [after (Gallerati et al., 2023)]

This integration revolutionizes well control engineering education and research by bridging theory and practice, as students engage with live operational data to interpret real-time anomalies and apply procedures under realistic conditions, cultivating data literacy and critical decision-making skills essential for modern digital rigs. It enables safe experimentation, allowing researchers to test novel well control algorithms (e.g., AI-driven kick detection) or high-consequence scenarios within the drilling simulators using the digital twin as a validation benchmark, eliminating risks associated with physical trials.

The system also facilitates human factors studies by analyzing crew performance, communication patterns, and stress responses during data-driven events, informing optimized training protocols and safety procedures. For academia, this approach modernizes curricula by replacing static lectures with adaptive, scenario-based learning, aligning education with industry's digital transformation and providing an unparalleled platform to develop future-ready engineers, advance research in complex well control challenges, and set new standards for safety and efficiency in drilling operations (Martins et al., 2025; Saputelli et al., 2024).

AI-ML Applications for Proactive Well Control

Artificial Intelligence (AI) is a broad field of computer science focused on creating machines that can perform tasks typically requiring human intelligence. This includes capabilities such as learning, problem-solving, perception, and language understanding. AI systems are designed to analyze data, identify patterns, and make decisions or predictions, often adapting their behavior over time based on new information.

Figure 4 illustrates a timeline of significant advancements in artificial intelligence (AI) and machine learning (ML), highlighting key developments that have shaped the field. The journey began in 1950 with Alan Turing's proposal of the Turing test, a foundational concept for evaluating machine intelligence. This was closely followed by the coining of the term 'Artificial Intelligence' in 1956 at the Dartmouth Summer Research Project, marking the formal inception of AI as a field of study (Anurag, 2001). The timeline progresses through the introduction of "Machine Learning" in 1959 demonstrating early attempts at human-computer interaction. The 21st century brought rapid acceleration that signify AI's increasing real-world impact and its move towards generative and highly capable models.

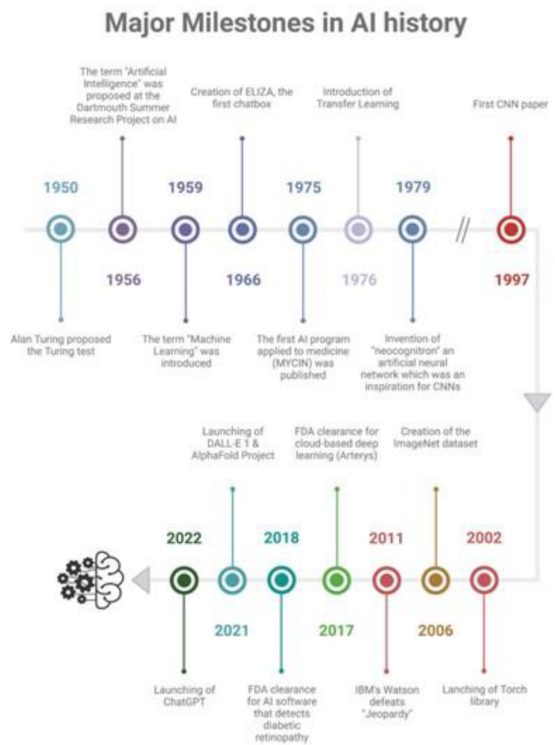


Figure 4—History of AI (Koltsakis et al., 2023)

At its core, AI's functionality is enabled by several foundational technological pillars, as illustrated in Figure 5. These core components include machine learning (ML), which empowers systems to learn patterns and make predictions from data without explicit programming; natural language processing (NLP), enabling machines to understand, interpret, and generate human language; computer vision, providing systems with the ability to interpret and analyze visual information from the digital world; and Robotics, integrating AI with physical mechanisms to enable autonomous or semi-autonomous interaction with the physical environment. Together, these elements form the bedrock of contemporary AI applications, driving innovation in domains ranging from automated decision systems to intelligent human-machine interfaces (Gamal, 2025).

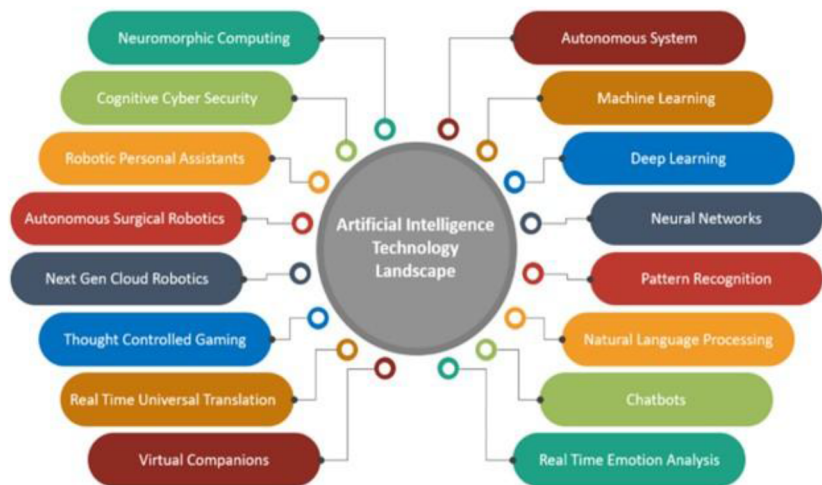


Figure 5—The core elements of AI (Aldoseri et al., 2023)

Machine learning (ML) is transforming well control and integrity management through predictive analytics and real-time monitoring. By processing vast operational datasets, ML models detect subtle anomalies and optimize drilling parameters, significantly enhancing safety and efficiency. Physics-based models rely on first principles (e.g., fluid dynamics, rock mechanics) to simulate well behavior, offering interpretability but requiring precise inputs (Figure 6). Data-driven models (e.g., ML) learn patterns from historical/real-time data, excelling in complex, nonlinear systems but needing robust datasets. Hybrid approaches combine strengths for enhanced accuracy.

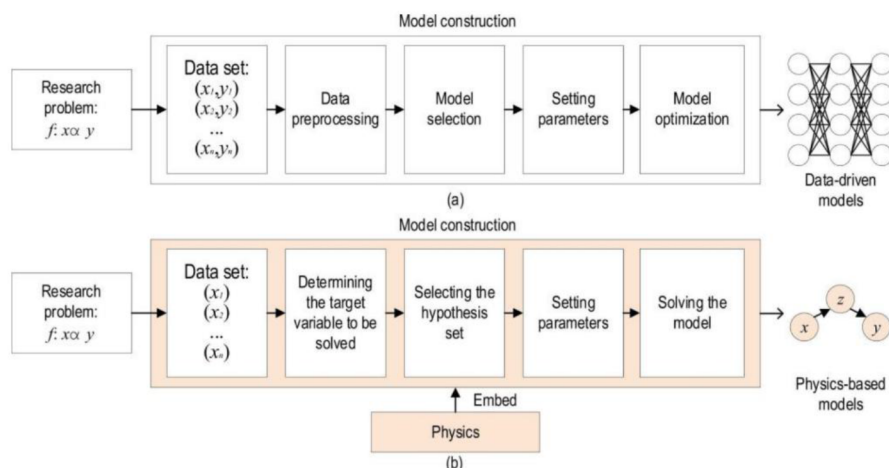


Figure 6—Models construction approach: (a) data-driven, and (b) physics-based [after (Wang et al., 2022)]

Figure 7 represents a structured data science workflow, progressing from business needs to technical execution and reporting (Gamal et al., 2023). It begins with business understanding to align objectives, followed by data understanding to assess available datasets. The data collection phase gathers raw inputs, which then undergo data preprocessing (cleaning/transformation) for analysis. Next, model building develops predictive solutions, and model evaluation validates their performance. Finally, reporting translates results into actionable insights. Notably, the workflow highlights data preprocessing and evaluation as iterative checkpoints, emphasizing their cyclical role in refining models.



Figure 7—Machine learning research for well control solutions (Gamal, 2025)

Early Kick Detection

Kicks that are uncontrolled influxes of formation fluids, pose severe risks if not detected promptly. Traditional methods monitor mud pit volume and flow rates but often lag in identifying micro-kicks. ML

models, trained on parameters like standpipe pressure, mud flow, and rate of penetration (ROP), detect subtle anomalies indicative of kicks. Supervised learning algorithms, such as multi-head self-attention networks, achieve high accuracy by analyzing gas evolution data and surface-riser-downhole monitoring systems. Recent studies demonstrate ML's superiority over conventional methods, with pilot-scale rig data validating its reliability for gas-kick detection (Obi et al., 2023; Zachopoulos and Kokkinos, 2023).

Pipe Stuck Prediction

Pipe sticking, a major cause of non-productive time (NPT), results from differential sticking or cuttings accumulation. ML models analyze torque, drag, mud properties, and ROP to predict sticking risks. Supervised learning classifiers categorize drilling states as "normal" or "at risk," while unsupervised methods like graph attention networks identify early warning signs in sparse datasets (Elahifar and Hosseini, 2022; Nautiyal and Mishra, 2022). Field tests of ML-driven stuck pipe detection systems (SPDS) across diverse wells confirm their efficacy in reducing NPT.

Wellbore Stability Prediction

Wellbore instability leads to hole enlargement and lost circulation. Traditional geomechanical models struggle with complex formations, whereas ML integrates drilling parameters, well logs, and rock properties to predict instability. Artificial neural networks (ANNs) excel in forecasting wellbore enlargement, while supervised models classify stability conditions. Physics-incorporated ML methods further refine predictions by coupling data-driven insights with mechanical principles, optimizing mud weight windows in real time (Cai et al., 2025; Ikobee et al., 2025; Liu et al., 2024).

Pore Pressure Prediction

Pore pressure uncertainties risk well control incidents. ML models integrate seismic, well log, and drilling data to predict pressures dynamically. Supervised algorithms like support vector machines reduce pre-drill analysis time from weeks to hours, with field studies showing <10% error margins in abnormally pressured reservoirs (Das and Maiti, 2024; Radwan et al., 2022).

Well Control Technology Innovations: Managed Pressure Drilling (MPD)

Drilling oil and gas wells involves a range of hazards that pose significant risks to personnel safety, environmental integrity, logistics, and operational costs. Over the past 160 years, drilling tools and systems have undergone continuous development, leading to the emergence of modern, increasingly automated drilling technologies. These advancements are now moving toward fully remote-operated drilling units. In parallel, industry-recommended practices have evolved to help standardize operations, promote environmental stewardship, and prioritize safety. One notable development is the formal definition of the well barrier envelope and its associated well barrier elements (WBEs).

The WBE was defined as a physical element that, in itself, does not prevent flow but, in combination with other WBEs, forms a well barrier envelope. A well barrier envelope is defined as a set of WBEs that encapsulate and prevent flow from a source. Identifying the well barrier envelopes is paramount before the commencement of an operational activity. The identified WBEs shall comply with their specific acceptance criteria and consist of:

1. Primary Well Barrier Envelope: is the first well barrier envelope that prevents undesired flow from a source of inflow/reservoir.
2. Secondary Well Barrier Envelope: is the well-barrier envelope that prevents undesired flow from a source of inflow/reservoir if the primary well-barrier fails.

In conventional drilling, the hydrostatic column of drilling fluid typically serves as the primary well barrier, providing overbalance against the maximum pore pressure of exposed formations. However, certain

formations, especially depleted or over-pressured zones, introduce elevated risks and, in some cases, render conventional drilling infeasible. Pressure-related hazards can lead to well control incidents, lost circulation, and stuck pipe events. Traditional open systems rely solely on increasing mud weight to manage these risks, which is often insufficient to stay within the narrow safe drilling margin (SDM), the pressure window between pore and fracture pressures. Maintaining well integrity in such environments requires the bottomhole pressure (BHP) to remain above pore pressure (to prevent influx) yet below fracture pressure (to avoid losses).

A pivotal industry advancement was the shift toward closed-loop drilling systems, enabled by the introduction of rotating control devices (RCDs). RCDs are drill-through devices that allow rotation of the drill string while maintaining pressure containment via dynamic seals. Initially implemented to mitigate surface fluid escape during pressure-challenged drilling, RCDs paved the way for more controlled drilling environments.

This innovation catalyzed the development of underbalanced drilling (UBD), a technique in which wellbore pressure is intentionally maintained below formation pressure. UBD required significant surface equipment, including but not limited to RCDs, multiphase separators, nitrogen production and compression units, and precision-controlled chokes. While UBD helped prevent reservoir damage and improved productivity, it introduced operational complexity and safety concerns, particularly in offshore environments, due to continuous surface flow. Its high capital expenditure (CAPEX) and the challenge of maintaining consistent underbalance further limited its widespread adoption.

To address these limitations, a safer, more controlled, and economical variation emerged: managed pressure drilling (MPD). MPD maintains the wellbore in a balanced or slightly overbalanced state with no intent to reach an underbalanced state. The objective is to precisely control the BHP within the SDM in real time, using specialized equipment and procedures. Unlike conventional methods that rely solely on mud weight adjustments, MPD enables near-instantaneous BHP manipulation via surface backpressure, allowing early and precise response to downhole pressure changes ([Gala and Ahead, 2011](#), p. 101).

Unlike conventional drilling, which reacts to kicks primarily by increasing mud weight, MPD systems provide real-time control over BHP, allowing dynamic responses to downhole pressure fluctuations without waiting for full-circulation methods to take effect. MPD redefines the primary well barrier to include not only the hydrostatic column but also applied surface backpressure, annular frictional losses, and all mechanical components of the closed-loop system, such as the RCD, MPD manifold, and choke system. The effective bottomhole pressure is thus expressed as:

$$\text{BHP} = \text{Hydrostatic Pressure} + \text{Annulus Friction Pressure} + \text{Annulus Back_Pressure}$$

In MPD, BHP adjustments can be made almost instantaneously, significantly mitigating the impact of well control events. MPD equipment is always considered part of the primary barrier envelope, independent of the secondary barrier envelope. The rig's Blowout Preventers (BOPs), rig chokes, and choke & kill lines remain the main well barrier elements within the secondary well barrier envelope. Kick indicators remain consistent across both conventional and MPD methodologies.

Early Kick Detection

Early kick detection remains a critical component of well control. In conventional operations, kicks are typically identified through signs such as increased return flow, pit volume gain, continued flow with pumps off, or deviations in pump pressure or speed. Real-time pressure while drilling (PWD) tools may also help detect inflow. However, these indicators can be delayed or obscured, particularly offshore, where large pit volumes and measurement limitations increase detection thresholds. Additionally, oil-based muds can dissolve formation fluids, delaying surface indications until gas comes out of solution, complicating response efforts.

Conventional well control models often assume constant fluid properties, overlooking the dynamic changes in mud weight and rheology that occur during drilling operations, particularly as drilled cuttings are generated and subsequently removed from the wellbore. As cuttings are transported to the surface and separated from the drilling fluid, the effective density and viscosity of the circulating fluid decrease. This results in changes in the effective BHP that are typically not accounted for in static kick tolerance or kill sheet calculations. Such variations can be critical, especially in wells with narrow pressure margins, where small decreases in hydrostatic pressure may lead to underbalanced conditions and formation influx.

Additionally, the removal of high-viscosity cuttings-laden fluid and its replacement with cleaner, lower-viscosity mud alters the equivalent circulating density (ECD), impacting the accuracy of pressure predictions. Failure to account for these dynamic changes in fluid properties introduces significant risk in well control planning and execution. Recognizing and integrating real-time rheological and density variations into pressure models is therefore essential for maintaining proper control over wellbore conditions (Houwen and Geehan, 1986).

MPD technology has evolved over the years from manual choke operation to hydraulically and electrically actuated systems, culminating in the deployment of fully automated MPD platforms. These systems offer closed-loop, algorithm-driven pressure control, enabling rapid and precise responses to dynamic well conditions. A key enabler of this evolution has been the integration of high-precision Coriolis flow meters. Introduced in MPD applications in the early 2000s, Coriolis meters provide continuous, highly accurate measurements of mass and volumetric flow rates, fluid density, and temperature. Their sensitivity significantly enhances early kick detection via monitoring differential flow rate and hence enables faster response to downhole changes.

As a practical application of the Coriolis effect, the Coriolis mass flow meter operating principle involves inducing a vibration of the flow tube through which the fluid passes (Figure 8). The vibration, though it is not entirely circular, provides a rotating reference frame that gives rise to the Coriolis effect (Emam et al., 2018). While specific methods vary according to the design of the flow meter, sensors monitor and analyze changes in frequency, phase shift, and amplitude of the vibrating flow tubes. The changes observed represent the mass flow rate and density of the fluid.

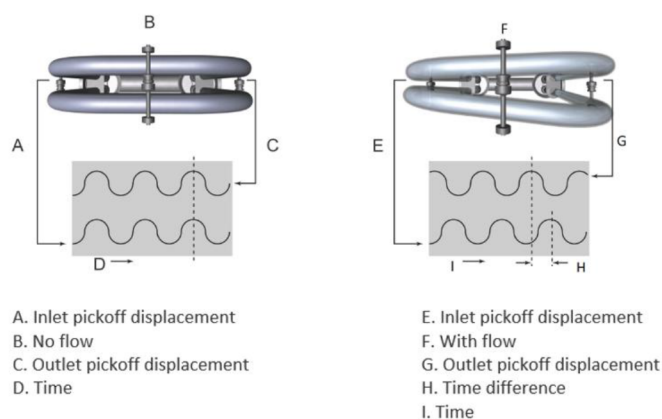


Figure 8—Coriolis Flow Meter measurement principle

The measuring tubes are forced to oscillate, producing a sine wave. At zero flow, the two tubes vibrate in phase with each other. When flow is introduced, the Coriolis forces cause the tubes to twist, resulting in a phase shift. The time difference between the waves is directly proportional to the mass flow rate. Volume flow rate is calculated from mass flow rate and the density measurement. The measuring tubes are vibrated at their natural frequency. A change in the mass of the fluid contained inside the tubes causes a corresponding change to the tube's natural frequency. The frequency change of the tube is used to calculate

density. Temperature is a measured variable that is available as an output. The temperature is also used internally in the sensor to compensate for temperature influences on Young's Modulus of Elasticity.

Traditionally, Coriolis flow meters used in MPD operations have been installed either downstream or, in some cases, upstream of the MPD chokes, where they exclusively measure the return flow from the annulus. In contrast, the flow of drilling fluid into the drill string is typically measured using conventional methods. These include mechanical or digital mud pump stroke counters that track the number of pump strokes. The fluid flow rate is then calculated volumetrically by multiplying the stroke count by the pump output per stroke, adjusted using a pump efficiency factor, commonly calibrated under unpressurized conditions.

In recent years, Coriolis flow meters have also been introduced to measure flow into the drill string; however, due to pressure rating constraints, these meters were installed upstream of the mud pumps. While this setup improved the accuracy of flow and density measurements compared to traditional stroke counting methods, it still neglected the effects of fluid compressibility induced by high-pressure conditions. As a result, both conventional and upstream Coriolis-based approaches do not fully capture the actual volumetric and density variations that occur under dynamic, pressurized drilling conditions.

High-Pressure Flow Meter

With the advent of high-precision instrumentation, a high-pressure flow meter (HPFM) utilizing the Coriolis effect has been introduced to the industry as an advanced technology for real-time drilling fluid characterization (Figure 9). Unlike traditional flow meters, the HPFM provides not only accurate volumetric and mass flow rates, fluid density, and temperature, but also real-time measurements of drilling fluid rheological properties. A key innovation in this system is its ability to be installed downstream of the mud pumps, prior to the fluid entering the drill string, thereby capturing the compressibility effects induced by frictional pressure losses throughout the drill string and annulus. This positioning significantly enhances the accuracy of hydraulic modeling. Furthermore, the continuous rheological data output by the HPFM offers substantial improvements in the performance of real-time engineering models within MPD modelling software platforms.

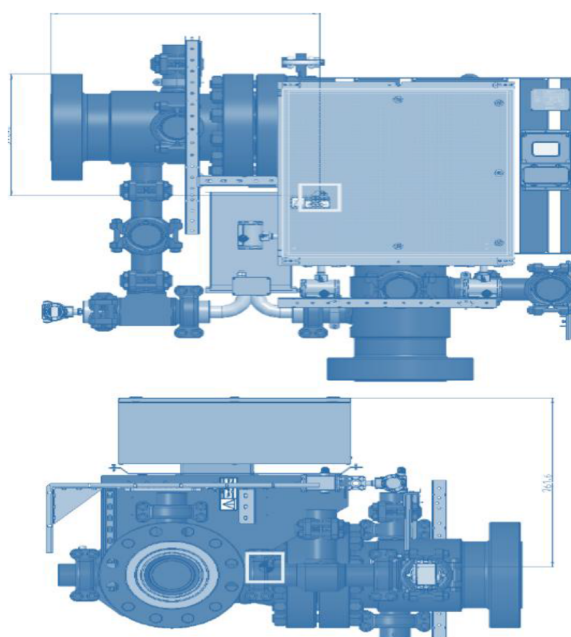


Figure 9—HPFM Technology

The HPFM operates based on a combined sensing architecture comprising a Coriolis mass flow meter, a differential pressure sensor, and a fixed choke (orifice restriction). As drilling fluid is pumped through

the HPFM manifold, it encounters the fixed choke, which creates a controlled restriction and generates a measurable pressure differential. Concurrently, the Coriolis sensor directly measures the mass flow rate. By integrating the differential pressure data with the Coriolis measurements, the system delivers highly accurate, high-resolution flow monitoring under high-pressure conditions. This dual-sensing methodology ensures precise flow control, which is critical for maintaining bottomhole pressure within the narrow operating window required in MPD applications.

Hydraulic Modelling

The integration of real-time hydraulic modeling into drilling operations has become a pivotal advancement in the evolution of MPD technology. As drilling environments grow increasingly complex, marked by depleted formations, narrow pressure margins, and extended-reach wells, the traditional approach of static, pre-drill hydraulic calculations proves inadequate. Real-time hydraulic modeling (RTHM) bridges this gap by dynamically simulating pressure and flow conditions along the wellbore, using live operational data to forecast downhole responses and proactively maintain BHP within a tight and safe drilling margin. In the context of MPD, RTHM (Figure 10) is particularly critical, as it allows precise control of ECD, detection of early influx or loss events, and optimization of choke settings to respond to even transient pressure variations caused by pump changes, pipe movement, or formation interactions.

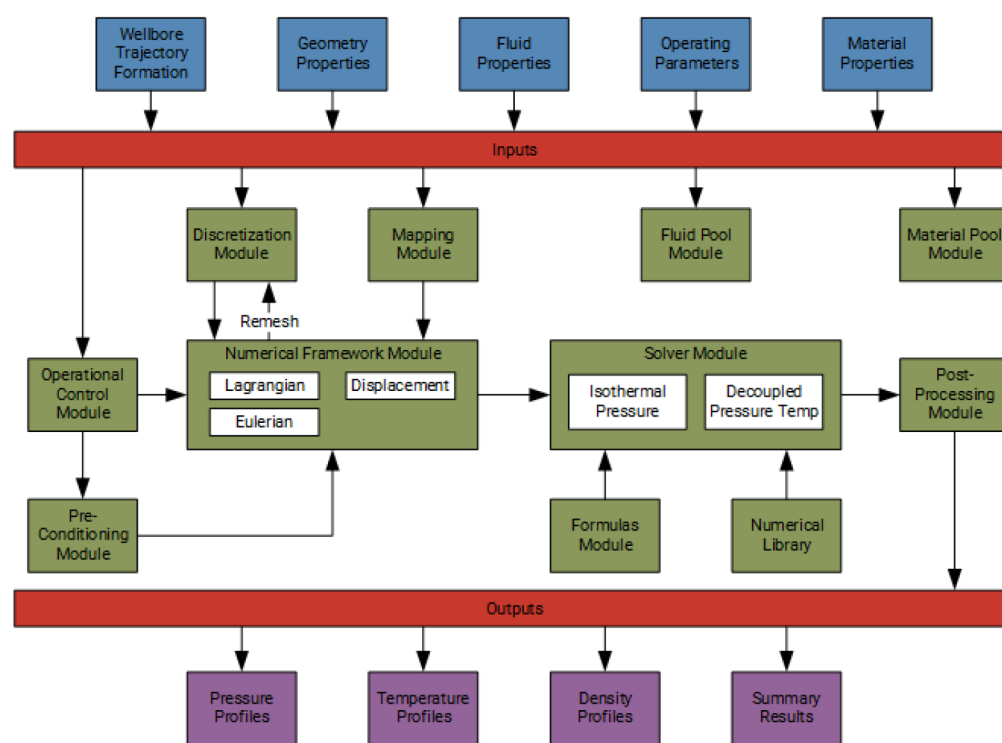


Figure 10—RTHM Workflow Diagram

Accurate hydraulic modeling requires continuous input from surface and potentially downhole sensors, providing high-resolution data on mud density, flow rate, temperature, annular pressure, and cuttings concentration. These inputs are often transmitted in real time through standard protocols such as WITS or WITSML, ensuring compatibility and synchronization between rig data systems and hydraulic modeling software. Sensors such as Coriolis flow meters deliver mass flow, density, and temperature data, while digital pressure sensors and pump stroke counters feed additional parameters into the modeling engine. For real-time integration, data must be time-aligned and of sufficient fidelity to enable fast computational cycling, allowing the model to react instantly to operational changes. Advanced edge computing platforms

often perform local data processing, presenting results as graphical trends that enable operators to monitor system dynamics intuitively, identify anomalies, and relay actionable insights to central control rooms or remote operations centers.

At the core of RTHM are mathematical models that represent the behavior of drilling fluids under different flow conditions and geometries. The selection of the appropriate hydraulic model is essential for simulation accuracy. The most common are the single-phase Newtonian and non-Newtonian models, two-phase flow models, and cuttings transport models. The Newtonian model assumes constant viscosity and is only suitable for simple, water-based fluids. In contrast, non-Newtonian models are used for most drilling muds, which exhibit shear-thinning behavior and variable viscosity as shown in Figure 11. Among these, the Bingham Plastic model assumes that the fluid has a yield stress that must be exceeded before flow begins, followed by a linear relationship between shear stress and shear rate. This model is widely used due to its simplicity and general applicability to many water-based mud systems. However, it can under- or overestimate hydraulic parameters in fluids with more complex behavior. The Power Law model accounts for shear-thinning but ignores yield stress, making it suitable for some oil-based and synthetic muds where flow starts instantly at any stress level. A more comprehensive model is the Herschel–Bulkley model, which combines yield stress with a non-linear shear rate–shear stress relationship. It provides greater accuracy over a wide range of mud systems, especially in transient flow conditions, but requires more computational effort and well-calibrated rheological inputs.



Figure 11—Data Entry for Drilling Fluid Properties – Necessary for RTHM

In real-time modeling, these fluid behavior models are embedded into flow and pressure simulations along the wellbore. The system calculates annular pressure losses, ECD, surge and swab effects, and cuttings transport efficiency. In MPD operations, these predictions are continuously updated to maintain BHP within the designated safe drilling envelope. Furthermore, two-phase flow models become relevant in cases of gas influx or when drilling with compressible fluids such as aerated mud or foam. These models require accurate prediction of slip velocities, void fractions, and flow regime transitions, adding further complexity to the hydraulic simulation. Meanwhile, cuttings transport models simulate the slip, accumulation, and removal

of drilled solids, which affect annular pressure and can lead to erratic BHP behavior if not accounted for in real time.

Real-time hydraulic modeling plays a foundational role in enabling MPD to meet its objective of maintaining precise pressure control throughout the drilling process. The integration of RTHM into MPD operations not only enhances operational awareness but also facilitates closed-loop control systems (Figure 12). These systems use discrepancies between measured and modeled parameters to auto-adjust surface backpressure or modify operational parameters in real time, improving precision in maintaining BHP. This enables automated control strategies that are essential for modern drilling environments and results in a substantial reduction in non-productive time (NPT), enhanced wellbore stability, and minimized risk of well control incidents.

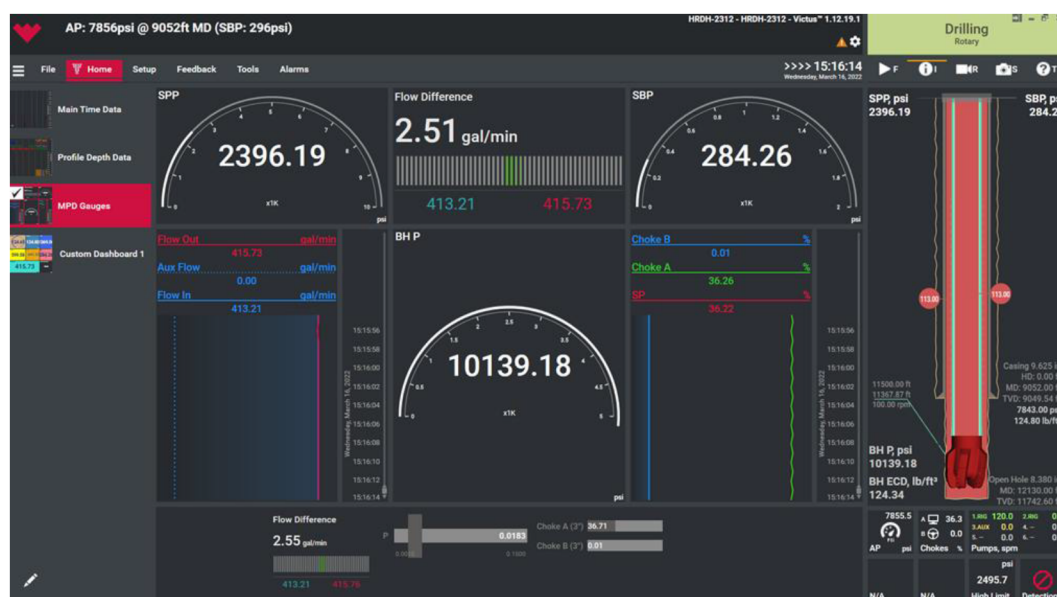


Figure 12—RTHM Integration within MPD platform

Additionally, the feedback provided by RTHM enables real-time adaptability. It allows for improved engineering workflows, such as real-time pore pressure/fracture gradient (PPFG) evaluation, mud program optimization, and predictive modeling of wellbore breathing or ballooning effects.

Case Studies

To reinforce the discussion on well control applications in drilling, representative cases are presented in this section. Each case draws on simulator outputs that demonstrate how digital twins, automation, and cyber-physical models improve competence development and safeguard well integrity.

Case 1: Well Control through Digital Twins

Well control training benefits directly from IR4's ability to couple physics-based models with immersive visualization. In this case, simulators replicate kick detection and containment workflows using downhole visualization interfaces (Figure 13), rig-floor instrumentation twins (Figure 14), and data-driven dashboards. Dedicated well control displays (Figure 15) reproduce influx events, tracking operator responses to generate competence metrics such as time to shut-in and choke stability. This case illustrates how digital twins provide information transparency, predictive analytics, and auditable evidence of well control competence, aligning with the core goal of enhancing well integrity. This represents how IR4 technologies enable visualization of hidden downhole states by fusing real-time sensor data with predictive models by presenting pore pressure,

fracture gradient, and thermal environment, the simulator enhances situational awareness critical for well integrity assurance (Figure 13).

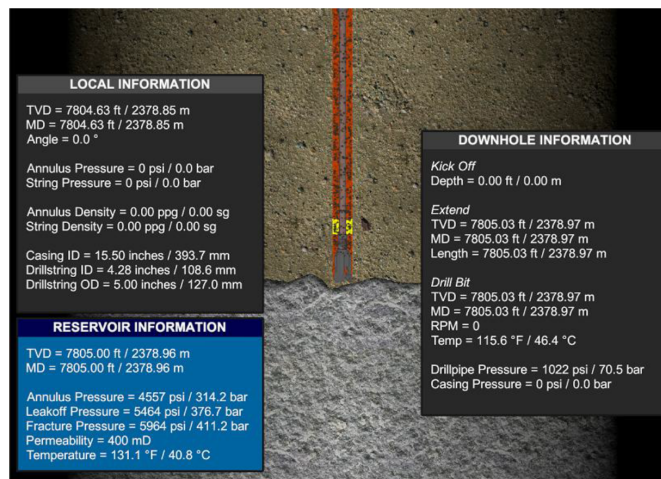


Figure 13—Digital twin of rig instrumentation, showing hook load, block position, brake pressures, and auto-driller settings.

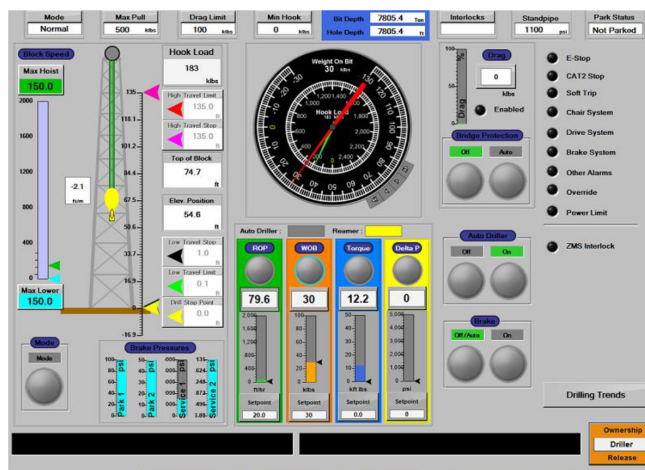


Figure 14—Amphiion-style rig control dashboard displaying drilling trends, ROP, block height, and flow balance.

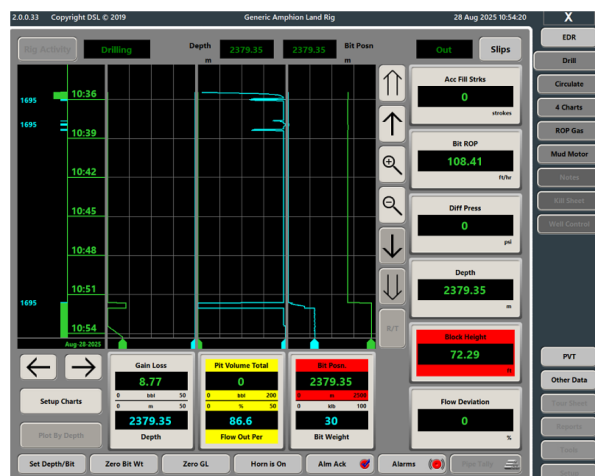


Figure 15—Simulator readout of well control data, including pressures, flow rates, and kick volume.

This digital dashboard reflects the shift from reactive surface indicators to predictive 4IR models. The coupling of big data analytics and real-time visualization supports early anomaly detection and proactive well control management (Figure 14).

Such simulators model kick progression and containment workflows, that emphasis on predictive ML and anomaly detection. Recording trainee responses builds a dataset for competence assessment and feeds future ML advisory systems Figure 15.

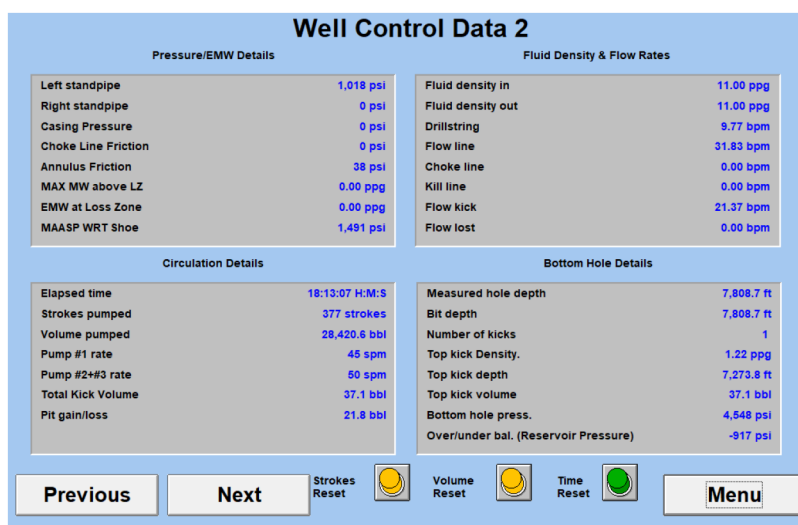


Figure 16—Well control simulator output with influx volume, flow imbalance, and kick detection trends.

Case 2: Managed Pressure Drilling (MPD) Automation

This case highlights the integration of MPD systems into IR4 frameworks. The MPD status screen (Figure 17) exemplifies adaptive automation, balancing pressure and flow in real time. The MPD control interface (Figure 18) shows how algorithmic choke and pump adjustments reduce human error and improve pressure management. The rotating control device (RCD) (Figure 19) anchors this system, enabling closed-loop annular flow control as part of a digital twin environment. Together, these elements demonstrate how intelligent MPD automation powered by IoT sensors, machine learning, and predictive control acts as a proactive barrier against influx and instability.

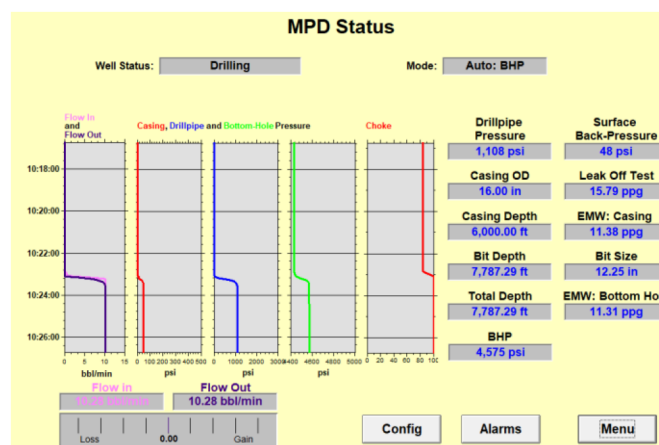


Figure 17—Managed Pressure Drilling (MPD) control interface showing real-time flow balance, pressure monitoring, and automated backpressure adjustments.

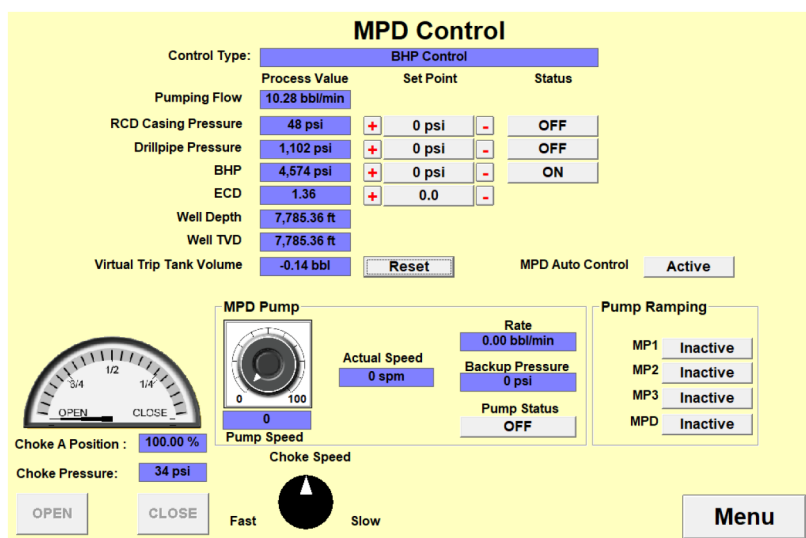


Figure 18—Automated MPD control panel with bottomhole pressure (BHP) set point, pump speed, and choke management.

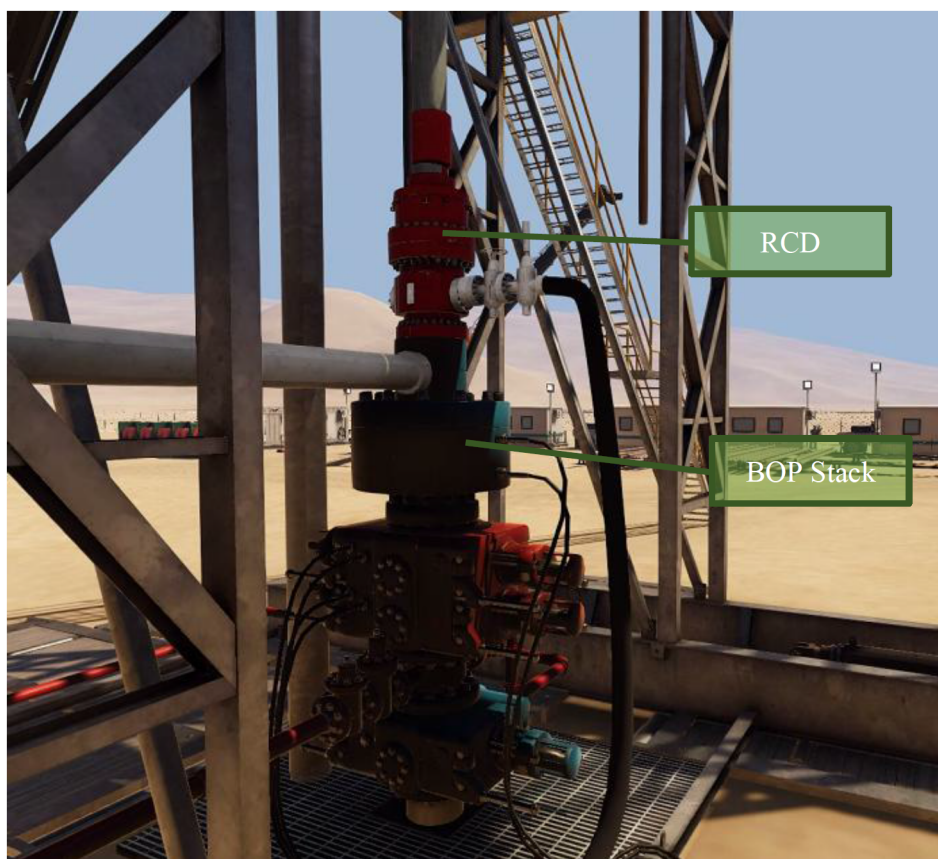


Figure 19—Digital simulation of a Rotating Control Device (RCD) used in Managed Pressure Drilling (MPD) operations.

Figure 18 illustrates the intelligent MPD automation framework. By embedding adaptive control algorithms, simulators teach operators how closed-loop systems minimize ECD excursions, enhancing safety and economic efficiency in high-risk wells.

The RCD is central to MPD workflows, enabling closed-loop pressure management by diverting annular returns through choke systems. Within a 4IR-enabled simulator, the RCD is modeled as part of the cyber-physical MPD automation framework, demonstrating how adaptive choke control, sensor integration, and

predictive algorithms maintain bottomhole pressure within the safe drilling window. Figure 19 highlights how digital twins prepare operators to manage influx and loss events while reinforcing well integrity.

Case 3: Stuck Pipe and Jarring Simulations

The case focuses on mechanical hazards, particularly stuck pipe. The jarring data interface (Figure 20) replicate the real-time force balances involved in freeing a stuck string. The stuck pipe simulation (Figure 21) reproduces differential sticking and pack-off conditions, allowing operators to train in anomaly recognition and corrective jarring techniques. This case demonstrates how cyber-physical drillstring models enhance experiential learning, reduce non-productive time, and prepare drillers for high-risk recovery operations.



Figure 20—Simulator display of hydraulic jar parameters and force balance during jarring operations

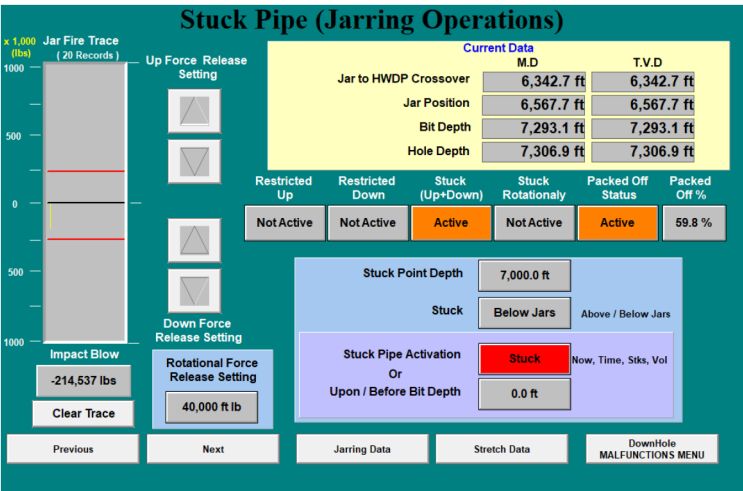


Figure 21—Virtual stuck pipe scenario with visualization of jarring force, depth, and activation conditions.

Across these three cases, the figures illustrate simulators as more than training aids: they function as 4IR-enabled digital ecosystems. Whether in well control, MPD automation, or stuck pipe remediation, simulators couple real-time data, physics models, and intelligent automation to improve decision-making and safeguard well integrity.

Figure 21 exemplifies how 4IR-enabled simulators reproduce operational hazards here, differential sticking and pack-off. Such digital training workflows allow drillers to practice anomaly recognition and remedial jarring techniques, generating auditable competence metrics (time to detection, pressure response).

Conclusions and Future Road Map

The integration of the fourth industrial revolution (4IR) technologies into well control and integrity management represents a transformative shift in the oil and gas industry's operational paradigm. This research has demonstrated that the convergence of machine learning, internet of things (IoT), big data analytics, robotics, automation, digital twins, and academic simulators not only addresses the limitations of conventional well control approaches but also establishes a proactive, predictive, and adaptive operational framework.

From the academic and industrial perspectives, the synergy between high-fidelity well control simulators and digital twin models enables a closed-loop system for both training and operational decision-making. Real-time integration of rig sensor data with physics-based and data-driven models fosters a deeper understanding of complex wellbore conditions, while providing a safe, risk-free environment to test innovative control strategies. The inclusion of academic simulators in this framework enhances workforce competence, bridges the gap between theoretical knowledge and field operations, and accelerates the adoption of advanced control methodologies such as AI-driven kick detection and managed pressure drilling (MPD).

Machine learning applications, specifically in early kick detection, pipe stuck prediction, wellbore stability, flux estimation, and pore pressure prediction, have proven capable of delivering measurable improvements in safety, non-productive time reduction, and drilling performance optimization. The shift from threshold-based monitoring to intelligent anomaly detection systems ensures that micro-events, often precursors to critical incidents, are captured and acted upon before escalating into operational hazards.

Managed Pressure Drilling, when augmented by AI and IoT-driven analytics, further exemplifies the potential of 4IR technologies to maintain precise downhole pressure control, particularly in narrow pressure window environments. Such integration reduces the margin for error, mitigates well control incidents, and optimizes drilling parameters dynamically.

Looking ahead, the continuous evolution of AI and ML models; particularly hybrid approaches that combine the interpretability of physics-based simulations with the adaptability of data-driven learning; will further enhance predictive accuracy and operational resilience. Collaborative industry-academia research will be essential in addressing challenges related to data standardization, model generalization across diverse geologies, and integration into real-time control systems.

Ultimately, the findings affirm that the future of well control and integrity lies in the seamless integration of intelligent technologies with operational workflows. By transitioning from reactive to proactive well management, operators can achieve enhanced safety, improved economic outcomes, and stronger environmental stewardship. The adoption of 4IR-driven well control frameworks will be critical in shaping the next era of sustainable, efficient, and intelligent drilling operations.

Acknowledgment

The authors gratefully acknowledge the research collaboration with Weatherford Company and the College of Petroleum Engineering & Geosciences at King Fahd University of Petroleum and Minerals through joint academic-industrial partnership.

References

1. Aldoseri, A., Al-Khalifa, K.N., Hamouda, A.M., 2023. Re-Thinking Data Strategy and Integration for Artificial Intelligence: Concepts, Opportunities, and Challenges. *Applied Sciences* **13**, 7082. <https://doi.org/10.3390/app13127082>
2. Anurag, A.S., 2001. *Early Beginnings of AI: The Field of Research in Computer Science*, in: <https://services.igi-global.com/resolvedoi/resolve.aspx?doi=10.4018/979-8-3693-2643-5.ch001>. IGI Global.
3. Cai, H., Yu, Y., Liu, Y., Gao, X., 2025. *Frontiers | Machine learning approach for prediction of safe mud window based on geochemical drilling log data*. <https://doi.org/10.3389/feart.2025.1529320>
4. Das, G., Maiti, S., 2024. A machine learning approach for the prediction of pore pressure using well log data of Hikurangi Tuaheni Zone of IODP Expedition 372, New Zealand. *Energy Geoscience* **5**, 100227. <https://doi.org/10.1016/j.engeos.2023.100227>
5. Elahifar, B., Hosseini, E., 2022. Machine learning algorithm for prediction of stuck pipe incidents using statistical data: case study in middle east oil fields. *Journal of Petroleum Exploration and Production Technology* **12**, 2019–2045. <https://doi.org/10.1007/s13202-021-01436-3>
6. Emam, S.S., Brand, P.R., Gabaldon, O.R., Vityk, M., Leithy, A.E., Shanab, M.A., Hamdy, B., Abdel-Moniem, M., El-Desouky, W., El-Azm, H.A., 2018. *Use of Managed Pressure Drilling Paved the Way to a Massive Gas Field Discovery in Egypt Western Desert: Case History*.
7. Gala, D., Ahead, J.S.T.T.W., 2011. Managed Pressure Drilling 101: Moving Beyond "It's Always Been Done That Way" [WWW Document]. *JPT*. URL <https://jpt.spe.org/managed-pressure-drilling-101-moving-beyond-its-always-been-done-way> (accessed 8.12.25).
8. Gallerati, P., Bori, L., Curina, F., Qushchi, A.T., 2023. *A Novel Approach to the Use of Digital Twins for Drilling Rigs Allows Better Operator Performance, While Achieving Best-in-Class Safety Standards Through Drilling the Well Virtually First*.
9. Gamal, H., 2025. *Technological Integration and Innovative Strategies Harnessing Artificial Intelligence for Operational Excellence [WWW Document]*. <https://services.igi-global.com/resolvedoi/resolve.aspx?doi=10.4018/979-8-3693-6447-5.ch009>. URL <https://www.igi-global.com/gateway/chapter/www.igi-global.com/gateway/chapter/359302> (accessed 5.29.25).
10. Gamal, H., Elkatatny, S., Al Gharbi, S., 2023. Rig Sensor Data for AI-ML Technology-Based Solutions: Research, Development, and Innovations. Presented at the *ADIPEC*, OnePetro. <https://doi.org/10.2118/216429-MS>
11. Houwen, O.H., Geehan, T., 1986. *Rheology of Oil-Base Muds*.
12. Ikobee, K., Nwamaioha, C., Wadike, S., Agbora, F., 2025. *A Machine Learning Approach to Predicting Wellbore Instability in the Niger Delta*.
13. Koltsakis, E., Klontzas, M.E., Karantanas, A.H., 2023. *What Is Artificial Intelligence: History and Basic Definitions*. https://doi.org/10.1007/978-3-031-25928-9_1
14. Li, G., Song, X., Tian, S., Zhu, Z., 2022. Intelligent Drilling and Completion: A Review. *Engineering* **18**, 33–48. <https://doi.org/10.1016/j.eng.2022.07.014>
15. Liu, P., Li, J., Chen, B., Yan, G., Lei, Q., Liang, L., Huang, Y., Zhao, H., Wang, G., Sun, M., 2024. *Digital Wellbore Stability Prediction with Machine Learning*.
16. Martins, A.L., Tobisawa, R.Y.I., da Silva, F.R.G., dos Santos, M.V., Mendes, G.L., Fernandes, A.A., Spoladore, I.P.B., Silva, V.G., 2025. *Lessons Learned from Operating a Simulation-Based Drilling Smart Twin: A 10-Year and Over 600 Drilled Wells Journey*.
17. Nautiyal, A., Mishra, A.K., 2022. Machine learning approach for intelligent prediction of petroleum upstream stuck pipe challenge in oil and gas industry. *Environment, Development and Sustainability* 1–27. <https://doi.org/10.1007/s10668-022-02387-3>

18. Obi, C.E., Falola, Y., Manikonda, K., Hasan, A.R., Hassan, I.G., Rahman, M.A., 2023. A Machine Learning Approach for Gas Kick Identification. *SPE Drill & Compl* **38**, 663–681. <https://doi.org/10.2118/215831-PA>
19. Radwan, A.E., Wood, D.A., Radwan, A.A., 2022. Machine learning and data-driven prediction of pore pressure from geophysical logs: A case study for the Mangahewa gas field, New Zealand. *Journal of Rock Mechanics and Geotechnical Engineering* **14**, 1799–1809. <https://doi.org/10.1016/j.jrmge.2022.01.012>
20. Said, M.M., Pilgrim, R., Rideout, G., Butt, S., 2022. *Theoretical Development of a Digital-Twin Based Automation System for Oil Well Drilling Rigs*.
21. Saputelli, L., Duran, J., Rivas, F., Casas, E., Ramos, L., Bravo, C., Chacon, A., Temizel, C., Mubarak, S., Chacon, J., Escorcia, A., Elred, M., Diaz, S., Yrigoyen, A., Moricca, G., Querales, M., Lopez, C., 2024. *Success Cases and Lessons Learned After 20 Years of Oilfield Digitalization Efforts*.
22. Sudhakar, R., 2023. *Robust Digital Twin Approach Towards Operational Excellence and Sustainability Goals*.
23. Wang, J., Li, Y., Gao, R.X., Zhang, F., 2022. Hybrid physics-based and data-driven models for smart manufacturing: Modelling, simulation, and explainability. *Journal of Manufacturing Systems* **63**, 381–391. <https://doi.org/10.1016/j.jmsy.2022.04.004>
24. Zachopoulos, F.N., Kokkinos, N.C., 2023. Detection methodologies on oil and gas kick: a systematic review. *International Journal of Oil, Gas and Coal Technology* **33**, 1–19. <https://doi.org/10.1504/IJOGCT.2023.130372>